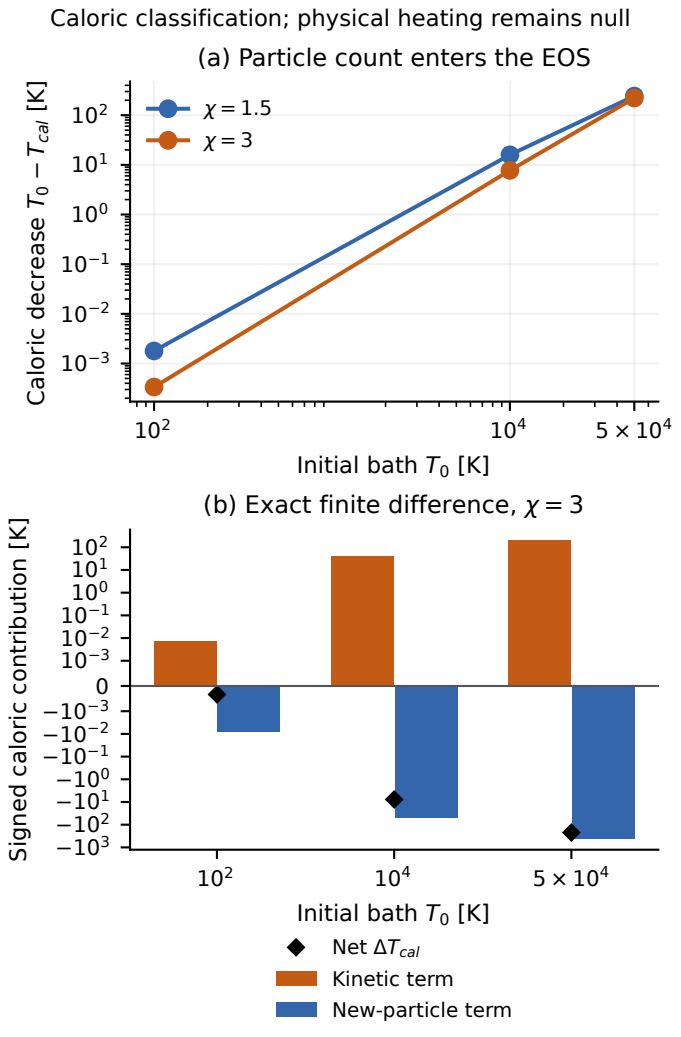
Caloric classification; physical heating remains null
(a) Particle count enters the EOS
$\chi = 1.5$
$\chi = 3$
Caloric decrease $T_0 - T_{cal}$ [K]
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
10^2
10^4
5×10^4
Initial bath T_0 [K]
(b) Exact finite difference, $\chi = 3$
Signed caloric contribution [K]
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
0
-10^{-3}
-10^{-2}
-10^{-1}
-10^0
-10^1
-10^2
-10^3
10^2
10^4
5×10^4
Initial bath T_0 [K]
Net ΔT_{cal}
Kinetic term
New-particle term